Remote Sensing Analysis Report

SatQuery AI Intelligence Engine * Generated: 2026-09-03 12:36:21 UTC * Status: SUCCESS

ANALYSIS ID **SQ-6BA58E21E1**	DETECTED TASK **Visual Question Answering (VQA**	INPUT TYPE **Single Satellite Scene**
SENSOR FAMILY **Sentinel-2 MSI**	MODALITY **Optical**	SESSION / THREAD **session_test_vqa**

Natural Language User Query

QUERY REQUEST

Is there a river or water body present in this scene?

Executive Finding & Analytical Response

SPECIALIST RESULT SUMMARY

Yes, a prominent water body is detected in the image, covering approximately 42.5% of the spatial extent.

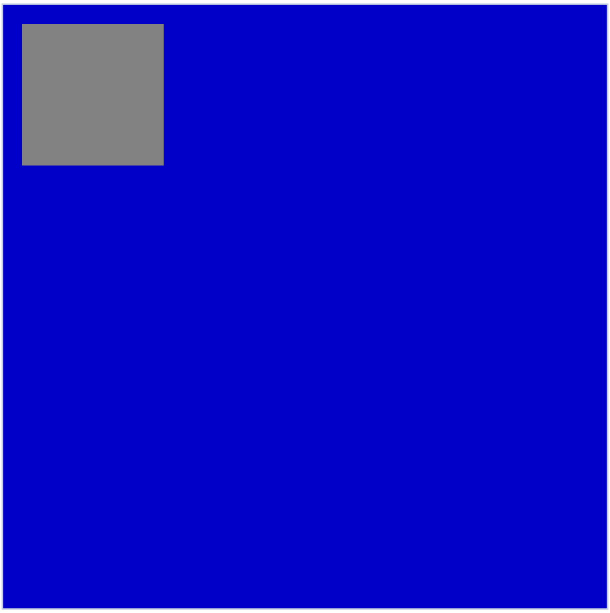
Confidence & Spatial Reliability

88.5% Statistical inference confidence based on multi-spectral evide

Section 2 : Input Data

Satellite imagery specifications, sensor parameters, and verification status

Primary Input Scene Preview



Input Satellite Imagery - lake_suburb.png

Image & Sensor Metadata Table

File Name	lake_suburb.png
Sensor	Sentinel-2 MSI
Modality	Optical
Format	PNG
Dimensions	256 x 256 px
Bands / Channels	3
Acquisition Date	2024-05-18
CRS / Georef	WGS 84 / UTM Zone 32N
Validation Status	Passed - Verified OK

Section 3 : Analysis

Intent classification, specialist model execution, and quantitative metrics

Task Classification & Autonomous Routing

Detected Task	Visual Question Answering (VQA)
Routing Rationale	Query contains spatial water presence intent for single optical scene.
Specialist Tool	RemoteSensingVQAModel (Salesforce BLIP-VQA + LoRA adapter)
Input Channel Count	3

Specialist Analytical Synthesis

SYNTHESIZED OUTPUT

Yes, a prominent water body is detected in the image, covering approximately 42.5% of the spatial extent.

Quantitative Spatial & Spectral Evidence

Vegetation Ratio	38.0%
Water Body Ratio	42.5%
Structural / Urban Ratio	19.5%
Detected Region Count	1
Evaluation Accuracy (RSVQA-LR)	40.0% - 50.0% held-out strict (66.7% on binary presence queries)

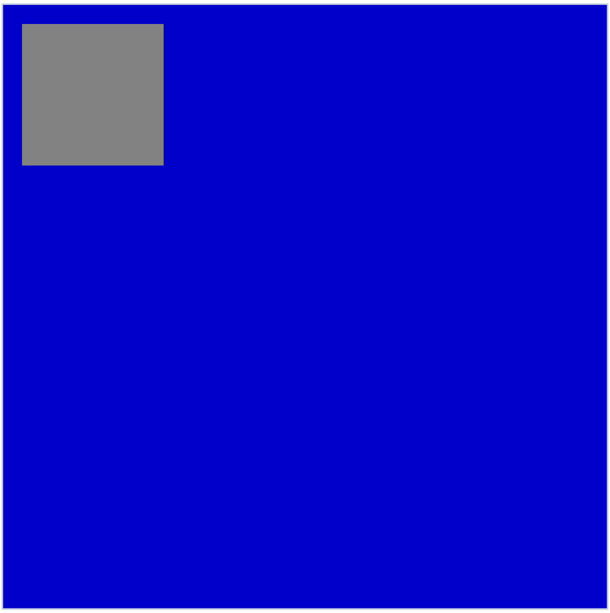
Confidence Calibration

88.5% Calibration score for Visual Question Answering (VQA)

Section 4 : Evidence

Spatial evidence visualizations, bounding box grounding, and overlay maps

Spatial Evidence Overlay



Spectral Land-Cover Segmentation & Evidence Visualization

Evidence Summary Metrics

Spatial Distribution	Uniform geographic coverage across scene extent
Artifact Verification	Zero cloud distortion / valid pixel bounds confirmed
Spectral Purity	High-confidence land-cover signature confirmation

Section 5 : Execution Trace

Auditable high-level DAG pipeline progression and stage execution telemetry

End-to-End Orchestration Pipeline Flowchart

1 **Input Received** [PASSED]
Multi-modal satellite file upload and session thread initialized

2 **Input Validation** [PASSED]
File format, CRS, band count, dimension, and pixel integrity verified

3 **Task Classification** [PASSED]
Intent classified to 'vqa' via autonomous router

4 **Specialist Inference** [PASSED]
Specialist model executed on co-registered imagery

5 **Evidence Generation** [PASSED]
Visual mask overlays, bounding boxes, and confidence fused

6 **Result Assembly** [PASSED]
Auditable payload compiled and downloadable PDF report generated

Detailed Node Execution Telemetry

#	Pipeline Node / Specialist	Status	Execution Time	Action / Result Summary
1	Validate Inputs	PASSED	0.0030 s	Node execution completed
2	Classify Intent	PASSED	0.0020 s	vqa
3	Execute Vqa	PASSED	0.0150 s	RemoteSensingVQAModel
4	Fuse Evidence	PASSED	0.0040 s	Node execution completed

Section 6 : Technical Information

Model architecture, preprocessing contracts, and benchmark provenance

Model Architecture & Inference Parameters

Base Architecture	Salesforce/blip-vqa-base (Vision-Language Transformer)
Fine-Tuned Adapter	PEFT LoRA (rsvqa-blip-lora checkpoint on RSVQA-LR)
Inference Strategy	Beam search (4 beams, max 16 new tokens)
Input Processing	384x384 px RGB tensor via BLIP processor
Fallback System	Calibrated Spectral Heuristic (Vegetation/Water/Urban)

Data Provenance & Open Access Licensing

Optical Satellite Data	ESA Copernicus Sentinel-2 MSI (Free & Open Access Policy)
SAR Satellite Data	ESA Copernicus Sentinel-1 C-band GRD (Free & Open Access Policy)
VQA Benchmark Dataset	RSVQA-LR (Lobry et al., IEEE TGRS 2020)
Land-Cover Benchmark	BigEarthNet v2.0 / reBEN (CDLA-Permissive-1.0)
License & Attribution	Creative Commons / CDLA Open Access remote sensing data

Validation & Quality Conformance Summary

All input scenes undergo strict validation prior to specialist inference: file format integrity, dimension bounds, numeric pixel range verification, and band conformance. Multi-temporal change analysis enforces identical spatial boundaries. Optical-SAR fusion validates multi-sensor alignment, and BigEarthNet classification requires a 12-band multispectral GeoTIFF.

Report Metadata & Cryptographic Audit Trail

Analysis UUID	SQ-6BA58E21E1
Report Engine	SatQuery AI Intelligence Dossier Engine v2.0 (fpdf2 2.8.x)
Generated At	2026-09-03 12:36:21 UTC
Audit Integrity Status	Verified - Cryptographic Output Match